Hyperparameters

For **Lasso Regression**, the most important hyperparameters are:

Hyperparameter	Common values	One-line explanation
alpha	`0.001, 0.01, 0.1, 1, 10, 100`	Controls L1 regularization strength; higher values takes the more feature selection more feature selection.
max_iter	`1000, 5000, 10000`	Maximum number of iterations allowed for training.

RIdge

Hyperparameter	Common values	One-line explanation
alpha	0.001, 0.01, 0.1, 1, 10, 100	Controls the strength of L2 regularization; higher values reduce overfitting more.
max_iter	1000, 5000, 10000	Maximum number of iterations allowed for training.

Elasticnet

Hyperparameter	Common values	One-line explanation
alpha	0.001, 0.01, 0.1, 1, 10, 100	Controls the overall regularization strength; higher values reduce overfitting more.
l1_ratio	0, 0.25, 0.5, 0.75, 1	Controls the mix of L1 and L2 regularization; 0 = Ridge, 1 = Lasso.
max_iter	1000, 5000, 10000	Maximum number of iterations allowed for training.

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`l1_ratio meaning`

- `l1_ratio=0  behaves like Ridge 0` → **`100% L2`** → `(0 = Lasso 100% Ridge)`
- `l1_ratio= 0.25` → **`25% L1(Lasso) + 75% L2(Ridge)`**
- `l1_ratio= 0.5` → **`50% L1(Lasso) + 50% L2(Ridge)`**
- `l1_ratio= 0.75` → **`75% L1(Lasso) + 25% L2(Ridge)`**
- `l1_ratio=1` → **`100% L1`** → `behaves like Lasso`

In **Elastic Net**, alpha controls **how strongly the model is regularized**.
Think of it as a **penalty knob**:

- alpha = 0 → **No regularization** → model can fit training data closely → higher risk of overfitting.
- alpha = 0.001 → **Very weak regularization**.
- alpha = 0.1 → **Moderate regularization**.
- alpha = 1 → **Stronger regularization**.
- alpha = 10 or 100 → **Very strong regularization** → coefficients are heavily reduced → may cause underfitting.

Simple example
ElasticNet(alpha=0.1, l1_ratio=0.5)
Here:

- **alpha=0.1** → overall strength of the penalty.
- **l1_ratio=0.5** → decides how that penalty is split between **L1 and L2**.

So remember:
alpha = "How much regularization?" l1_ratio = "What type of regularization (L1 vs L2)?"